

POWER ELECTRONICS

Power Parameter
Calculations

ROOT MEAN SQUARE

Calculations

General Equation for RMS Value

- RMS Definition

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

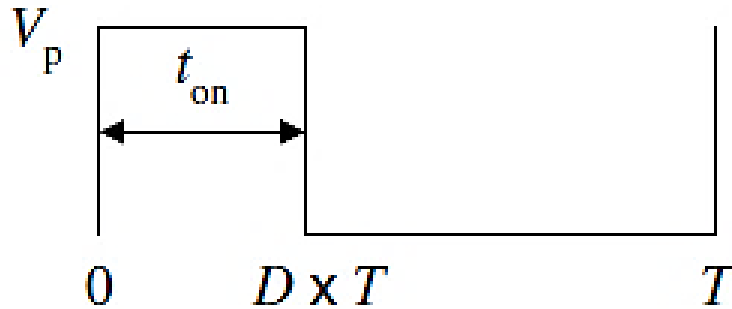
RMS Value of DC Waveform

$$v(t) = V_p$$

$$V_{rms} = \sqrt{\frac{1}{T} \int_0^T v^2(t) dt}$$

RMS Value of Rectangular Waveform

- For rectangular wave:



$$v(t) = \begin{cases} V_p & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

$$V_{\text{rms rectangle}} = \sqrt{\frac{1}{T} \left(\int_0^{DT} V_p^2 dt + \int_{DT}^T 0^2 dt \right)}$$

Example 4

- Calculate the rms value of a rectangular wave with the following parameters:

Frequency = 100 kHz

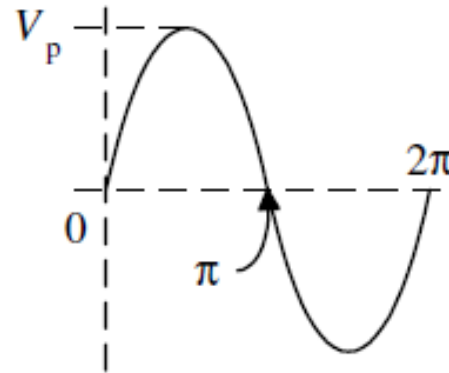
Duty cycle = 35%

$V_P = 15 \text{ V}$

RMS Value of Sinusoids

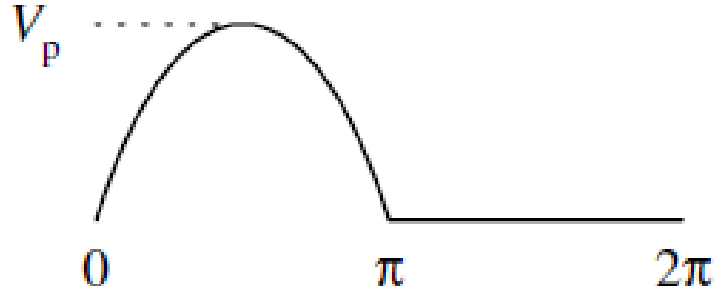
- RMS value of a sinusoid is 0.707 times the peak value

$$V_{\text{rms sine}} = \sqrt{\frac{1}{2\pi} \int_0^{2\pi} (V_p \sin \theta)^2 d\theta}$$



RMS Value of Half-Wave Rectified Sine

- For half wave rectifier



$$v(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ 0 & \pi \leq \theta < 2\pi \end{cases}$$

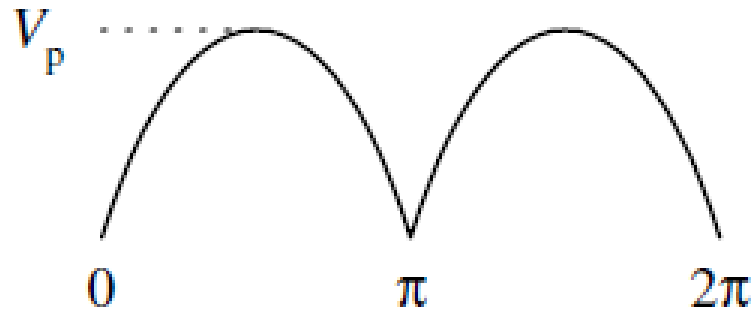
$$V_{\text{rms half sine}} = \sqrt{\frac{1}{2\pi} \int_0^{\pi} (V_p \sin \theta)^2 d\theta}$$

Example 5

- Calculate the rms value of a half-wave rectified sine wave with $f = 60$ Hz and $V_p = 170$ V

RMS Value of Full-Wave Rectified Sine

- Equal to the rms value of sinusoidal signal

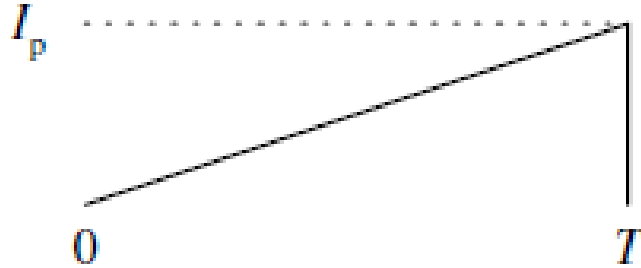


$$v(\theta) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ -V_p \sin \theta & \pi \leq \theta < 2\pi \end{cases}$$

$$V_{\text{rms full-wave rect}} = \sqrt{\frac{1}{2\pi} \left[\int_0^{\pi} (V_p \sin \theta)^2 d\theta + \int_{\pi}^{2\pi} (-V_p \sin \theta)^2 d\theta \right]}$$

RMS Value of Triangular Wave

- For ramp signal



$$i(t) = \frac{I_p}{T} t$$

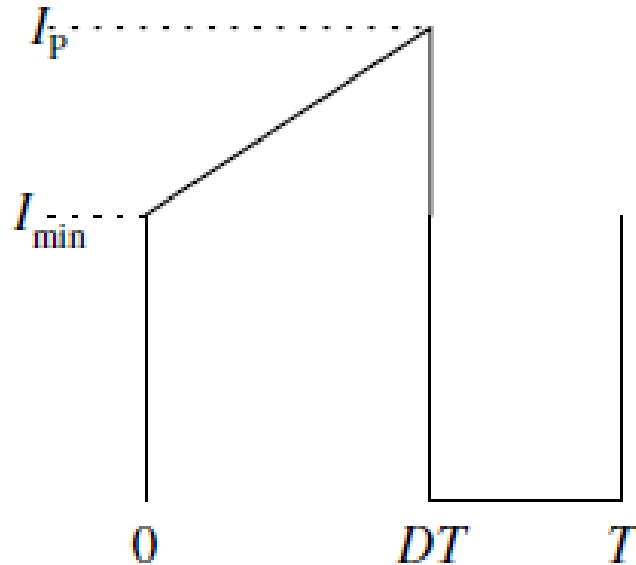
$$I_{\text{rms ramp}} = \sqrt{\frac{1}{T} \int_0^T \left(\frac{I_p}{T} t \right)^2 dt}$$

Example 6

- Calculate the rms value of a ramp with $I_p = 3 \text{ A}$, $f = 100 \text{ kHz}$.

RMS Value of Trapezoidal Wave

- Little bit complicated but can be solved with better understanding



$$i(t) = \begin{cases} \frac{I_p - I_{\min}}{DT} t + I_{\min} & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

$$I_{\text{rms trapezoid}} = \sqrt{\frac{1}{T} \int_0^{DT} \left(\frac{I_p - I_{\min}}{DT} t + I_{\min} \right)^2 dt}$$

POWER

Calculations

Instantaneous Power

- Need time domain equations of voltage and current

$$P_{inst} = v(t) \times i(t)$$

Average Power

- Integrate instantaneous power over a complete cycle

$$P_{\text{ave}} = \frac{1}{T} \int_0^T v(t) \times i(t) dt$$

DC Voltage and Current

- The simplest case

$$v(t) = V_p$$

$$i(t) = I_p$$

$$P_{\text{ave}} = \frac{1}{T} \int_0^T v(t) \times i(t) dt$$

Sinusoids

- For voltage and current that are in phase

$$v(\theta) = V_p \sin \theta$$

$$i(\theta) = I_p \sin \theta$$

$$P_{\text{ave}} = \frac{1}{T} \int_0^T v(t) \times i(t) dt$$

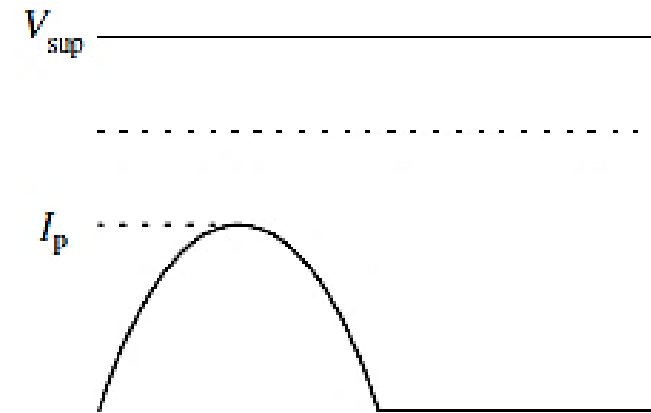
- Peak value and rms value

Example 1

- Calculate the average power for the following waves.

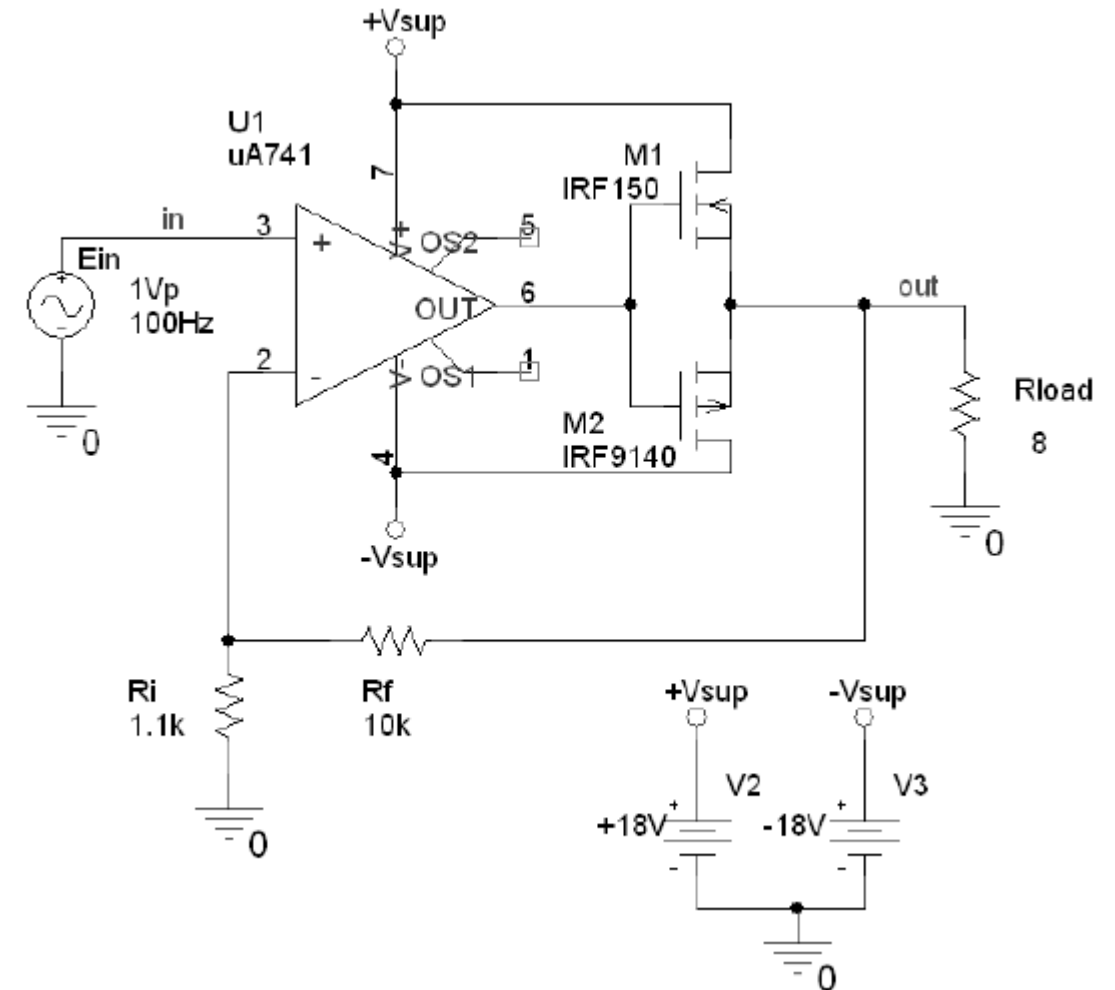
$$v(\theta) = V_{\text{sup}}$$

$$i(\theta) = \begin{cases} I_p \sin \theta & 0 \leq \theta < \pi \\ 0 & \pi \leq \theta < 2\pi \end{cases}$$



Example 2

- Calculate the power delivered to the load and the power provided by each supply.

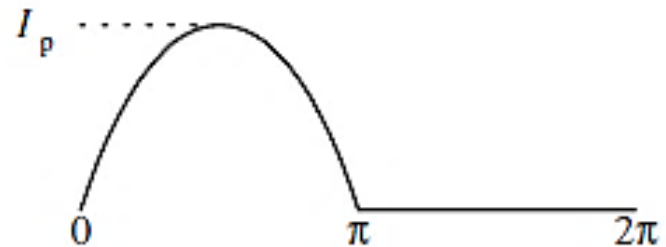
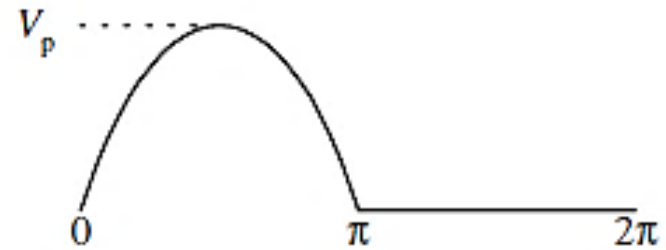


Half-Wave Rectified-Sine Wave

- Do yourself

$$v(t) = \begin{cases} V_p \sin \theta & 0 \leq \theta < \pi \\ 0 & \pi \leq \theta < 2\pi \end{cases}$$

$$i(t) = \begin{cases} I_p \sin \theta & 0 \leq \theta < \pi \\ 0 & \pi \leq \theta < 2\pi \end{cases}$$



Example 3

- Calculate the power delivered from a 170 V_P half-wave rectified sine voltage to a $51\ \Omega$ resistor.

Full-Wave Rectified-Sine Wave

- Derive the equation for the power delivered to a resistive load from a full-wave rectified sine.

Rectangular Wave

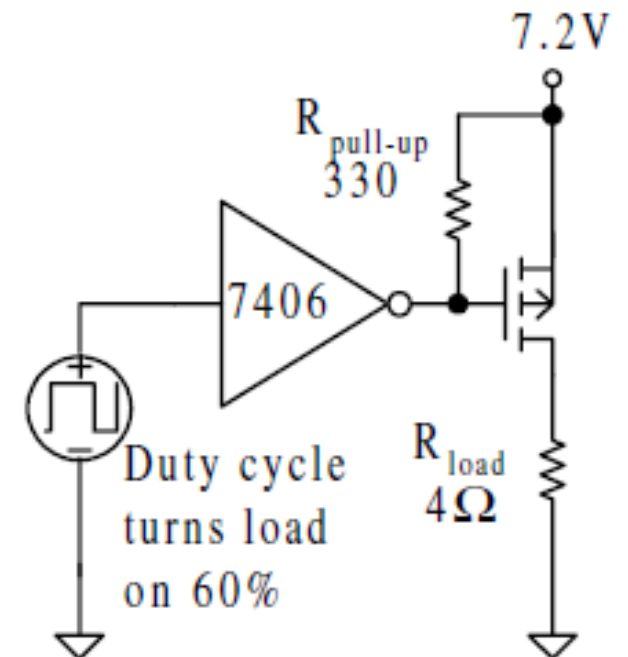
- Similar shape of current when rectangular voltage is applied across a resistive load.

$$v(t) = \begin{cases} V_p & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

$$i(t) = \begin{cases} I_p & 0 \leq t < DT \\ 0 & DT \leq t < T \end{cases}$$

Example 4

- For the circuit shown in figure, calculate the power dissipated by the load and by the transistor. When the pulse is high, the transistor's resistance is $0.2\ \Omega$



Reference

Power Electronics, Principles and Applications
J. Michael Jacob

Power Electronics
Ned Mohan